

TEJA VENKAT KUNDEM

Backend Software Engineer | Python · Django · FastAPI | REST APIs & Distributed Systems

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PROFESSIONAL SUMMARY

Backend Software Engineer with 1+ year of experience building production real-time AI systems and scalable backends in Python, Django, DRF, and FastAPI. Designed and shipped Aayu, a real-time speech-to-speech voice companion — owning low-latency voice pipelines (VAD → STT → LLM → TTS), multi-provider LLM orchestration (OpenAI Realtime, Anthropic Claude on AWS Bedrock), and async post-session processing over LiveKit/WebRTC. Skilled in microservices, distributed systems, LLM application development (LangChain/LangGraph), web scraping and browser automation, and cloud deployment on AWS with Docker, Kubernetes, and CI/CD.

CORE COMPETENCIES

REST API Design · Microservices · Distributed Systems · API Optimization · Database Design · System Design · OAuth 2.0 / JWT · LLM Application Development · CI/CD & DevOps · Agile/Scrum

TECHNICAL SKILLS

Languages: Python, SQL

Backend: Django, Django REST Framework (DRF), FastAPI, Celery

AI & LLM: OpenAI API, Anthropic Claude API, LangChain, LangGraph, LangSmith, LangFlow, Hugging Face, Scikit-learn, LLM Application Development, AI Backend Development

Databases: PostgreSQL, MongoDB, Redis, SQLite

Web Scraping & Automation: Playwright, Selenium, BeautifulSoup, Scrapy, Requests, curl-cffi, Kameleo, Browser Automation, Proxy Management

APIs & Authentication: REST APIs, OAuth 2.0, JWT Authentication, Webhooks, JSON

Cloud & DevOps: Docker, Docker Compose, Kubernetes, AWS (EC2, S3), Nginx, Gunicorn, Linux, CI/CD, GitHub Actions

Version Control: Git, GitHub

Architecture & Practices: Microservices, Distributed Systems, Scalability, System Design, Message Queues, Design Patterns, MVC Architecture, Clean Code, Performance Optimization, Debugging, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer — Gamyam Info Tech LLP

Aug 2025 – Present

Python · Django · DRF · LiveKit · OpenAI Realtime · Claude (Bedrock) · Celery · AWS

Andhra Pradesh, India

- Owned the end-to-end backend for Aayu, a real-time speech-to-speech voice companion — LiveKit (WebRTC) transport, LLM orchestration, and async post-session processing.
- Shipped 10+ production REST APIs in Django/DRF for high-volume traffic; cut API response time 25–30% via query optimization, indexing, connection pooling, and Redis caching.
- Engineered a low-latency voice pipeline (VAD → STT → LLM → TTS) using streaming TTS, provider pre-warming, and parallelized sidecars to cut perceived response time.
- Built a dual-agent architecture pairing an on-path voice model (OpenAI gpt-realtime) with an off-path Claude (Bedrock) reasoner, plus a safety classifier for distress detection across English, Hindi, and Telugu.
- Designed a microservices architecture with clear service boundaries, using PostgreSQL and MongoDB for structured and unstructured data.
- Optimized LLM cost and latency with Anthropic prompt caching and a config-driven “voice stack” for A/B rotation of model/voice/TTS combinations.
- Secured APIs with OAuth 2.0 / JWT and RBAC; containerized with Docker Compose, automated CI/CD via GitHub Actions, and deployed to AWS (EC2, S3) with per-turn cost/latency telemetry.
- Delivered backend microservices for Practo client platforms (Annual Health Checkup and Diagnostics): booking and order-lifecycle APIs, lab integrations, HMAC-SHA256 webhooks, and slot-lock reliability.

AI Engineer Intern — Gamyam Info Tech LLP

May 2025 – Jul 2025

Python · Scikit-learn · ML Pipelines

Hyderabad, India

- Developed and trained ML models in Python using Scikit-learn, applying preprocessing, feature engineering, and hyperparameter tuning.
- Integrated ML inference into backend services via clean REST interfaces, enabling product teams to consume predictions with minimal coupling.

PROJECTS

Aayu — Real-Time Speech-to-Speech AI Voice Companion

Python · AssemblyAI · OpenAI Realtime · Hume AI · Celery · Redis · PostgreSQL · S3

- Engineered a hybrid STT pipeline pairing AssemblyAI (transcription tuned for Indian-accented English) with OpenAI Realtime (prosody-aware understanding), raising accuracy while preserving emotional signal.
- Built a multi-layer emotion system: live prosody-driven tone adaptation, a mood state machine (–3 to +3), and post-session analytics via the Hume AI batch API, stored per-turn in PostgreSQL.
- Implemented a “backchannel” latency mask that plays a pre-cached acknowledgment while the real reply generates, turning dead air into an engaged pause.
- Architected async post-session processing with Celery + Redis: emotion analysis on S3 recordings and an LLM-as-judge loop generating per-user coaching for the next call.

Practo — Corporate Annual Health Checkup (AHC) Platform

Python · Django · DRF · Celery · Redis · HMAC · SSO

- Built the booking and order-management backend with Django and DRF, exposing RESTful APIs for the corporate annual health-checkup workflow.
- Integrated third-party diagnostic partners (Thyrocare, Healthians) behind a pluggable adapter architecture for provider-agnostic onboarding.
- Processed orders and notifications asynchronously with Celery and Redis to keep APIs responsive under load; secured partner and user access with HMAC request signing and SSO; automated PDF report generation and notification delivery workflows.

Practo — Diagnostics (DX) Platform

Python · Django · DRF · Celery · Redis · AWS SQS · Elasticsearch · Kubernetes

- Developed the core backend APIs powering diagnostic test booking, lab-catalog management, order-lifecycle tracking, and report delivery.
- Built integrations with partner laboratories for sample-collection scheduling, real-time order-status updates, and report retrieval.
- Optimized catalog and test search with Elasticsearch; orchestrated asynchronous workflows with Celery, Redis, and AWS SQS to decouple booking, processing, and delivery; deployed and maintained services in a Kubernetes-based production environment.

EDUCATION

B.Tech, Computer Science & Engineering — BVC Engineering College | CGPA: 8.10

2021 – 2025